COMPASS_TEMPEST_SGW_2022

Stephanie J. Wilson and Zoe Read

Sys.date

Link to 2021-2022 Sampling Protocol: https://docs.google.com/document/d/1YyvW0XtxktHx-h2cG5YStUpkXtn8LDbx/edit

##Set Up

Read in first data file

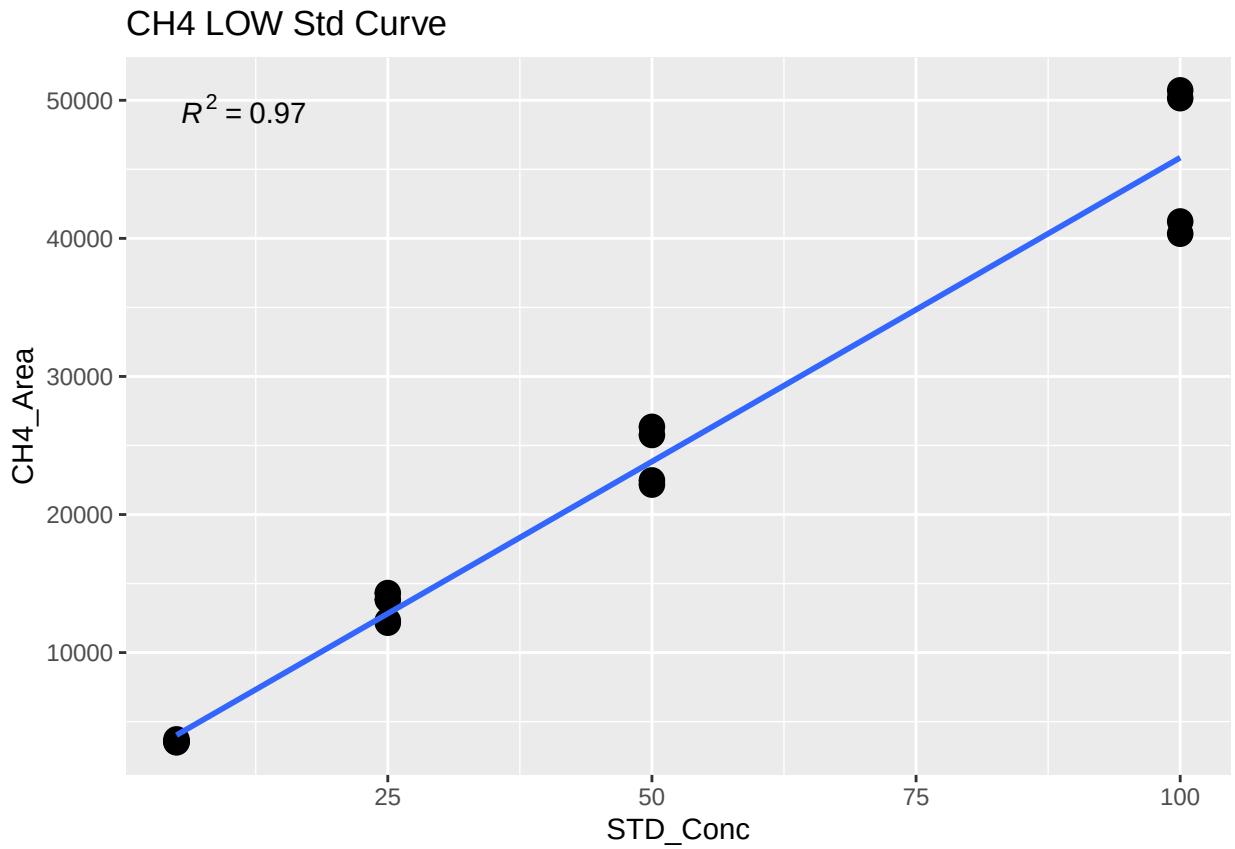
```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘across(c(STD_Conc, CO2_Area, CH4_Area), as.numeric)‘.
## Caused by warning:
## ! NAs introduced by coercion
```

```
## Warning: Expected 3 pieces. Additional pieces discarded in 2 rows [11, 86].
```

```
## Warning: Expected 3 pieces. Missing pieces filled with ‘NA‘ in 456 rows [54, 55, 129,
## 130, 192, 193, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214,
## 215, ...].
```

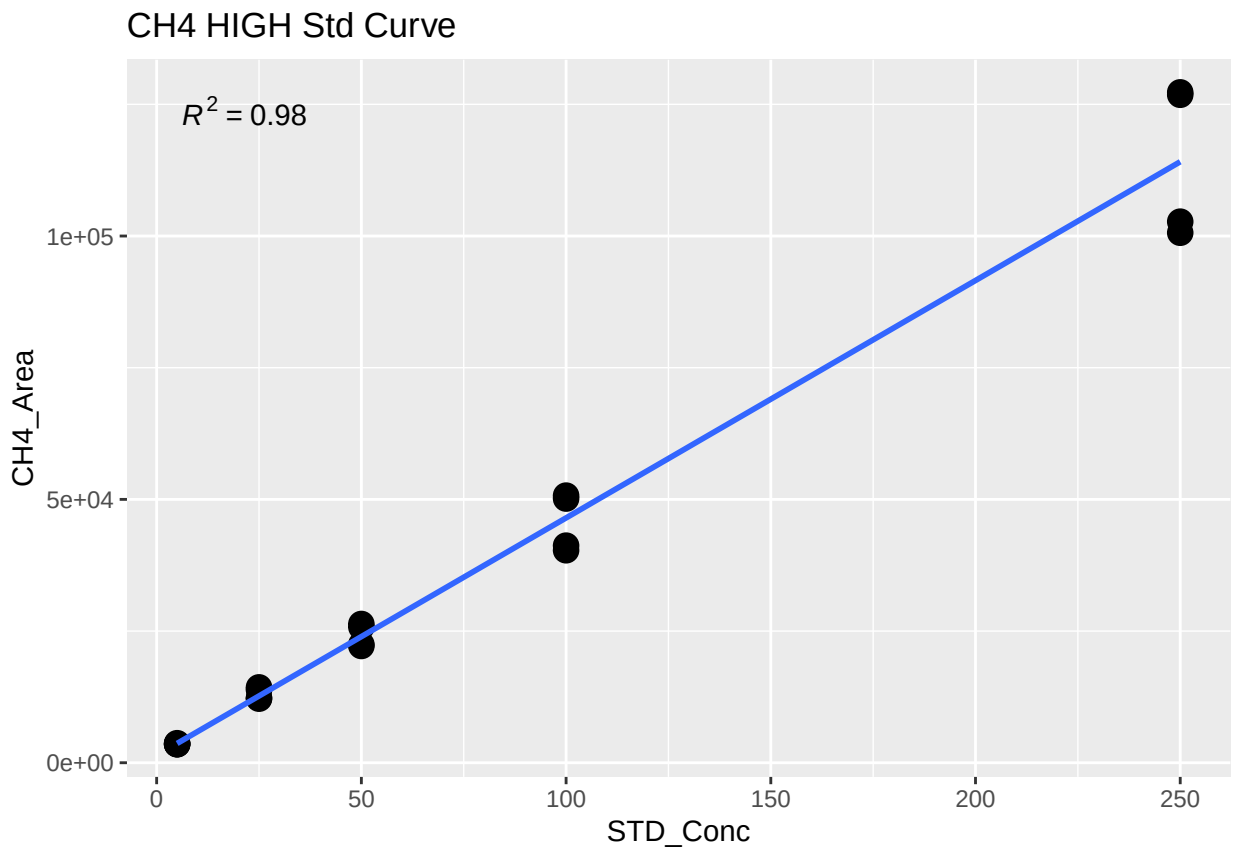
Assess standard curves

```
## ‘geom_smooth()‘ using formula = ’y ~ x’
```



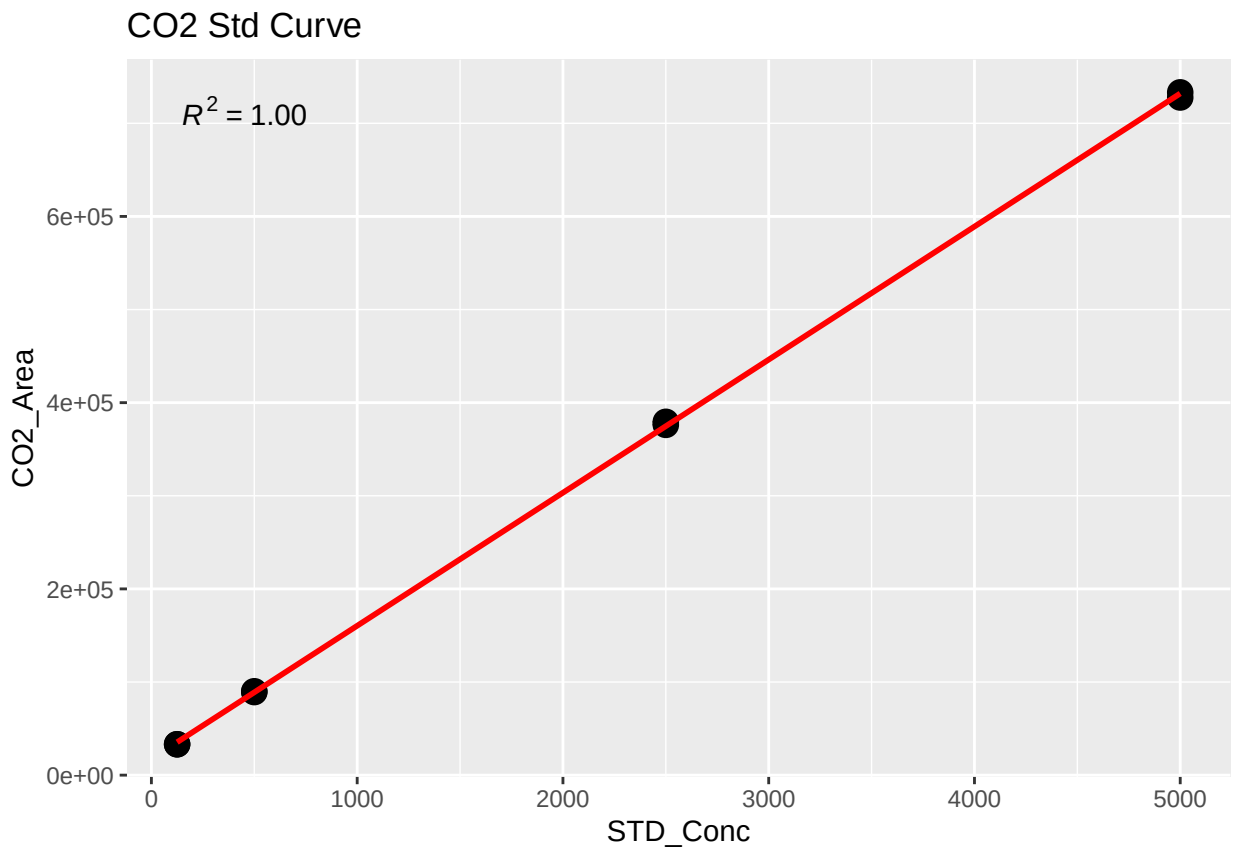

```
##
## Call:
## lm(formula = stds_ch4_low$CH4_Area ~ stds_ch4_low$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5507.6  -837.4  -472.4  1594.6  4876.4
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1820.70    1148.33   1.586    0.135
## stds_ch4_low$STD_Conc   440.24      20.03  21.981 2.98e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2847 on 14 degrees of freedom
## Multiple R-squared:  0.9718, Adjusted R-squared:  0.9698
## F-statistic: 483.2 on 1 and 14 DF,  p-value: 2.978e-12

## `geom_smooth()` using formula = 'y ~ x'
```



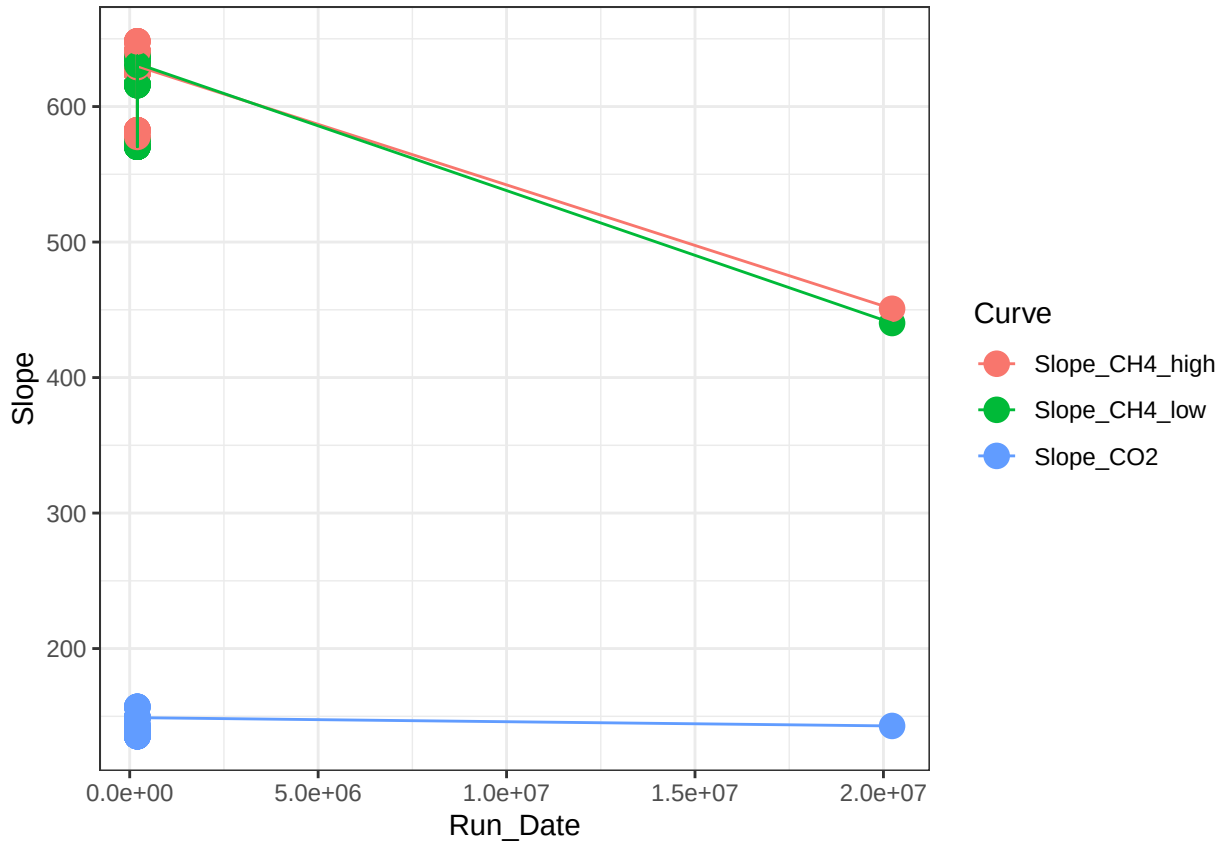

```
##
## Call:
## lm(formula = stds_ch4$CH4_Area ~ stds_ch4$STD_Conc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13481.0  -1555.3   -116.2   1955.4  13179.0
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1411.86    2035.17   0.694    0.497
## stds_ch4$STD_Conc   450.76      16.55  27.244 4.38e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6507 on 18 degrees of freedom
## Multiple R-squared:  0.9763, Adjusted R-squared:  0.975
## F-statistic: 742.2 on 1 and 18 DF,  p-value: 4.385e-16

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = stds_co2$CO2_Area ~ stds_co2$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4322.2 -2185.6   693.1  1287.3  4683.9
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)       1.751e+04  1.555e+03   11.26 2.93e-05 ***
## stds_co2$STD_Conc 1.429e+02  5.538e-01  258.03 2.29e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3035 on 6 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 6.658e+04 on 1 and 6 DF,  p-value: 2.287e-13

##   X          Curve        R2    Slope  Intercept Run_Date
## 1 1  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 2 2 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 3 3      Slope_CO2 0.9999951 138.7384 5556.86838   202304
## 4 4  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 5 5 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 6 6      Slope_CO2 0.9999951 138.7384 5556.86838   202304
```



Now calculate the CH4 & CO2 concentrations in ppm

```
#head(raw)

#pull out samples
Samples <-  raw %>%
  filter(str_detect(Sample_ID, "Control|FW|SW")) %>%
  select(Machine, Run_Date, Sample_ID, Plot, Grid_Square, Depth, Dilution_Factor, CO2_Area, CH4_Area, La
#head(Samples)

#Now flag any areas that are above the 100ppm area for CH4
Samples$CH4_Curve <- ifelse(Samples$CH4_Area > 50721, "High", "Low")
#head(Samples)

#Calculate CH4 concentrations in ppm
Samples$CH4_Conc_ppm <- ifelse(Samples$CH4_Area > 50721, (Samples$CH4_Area-Slope_CH4_high$Intercept)/Sl

#Calculate CO2 concentrations in ppm
Samples$CO2_Conc_ppm <- ((Samples$CO2_Area-Slope_CO2$Intercept)/Slope_CO2$Slope)

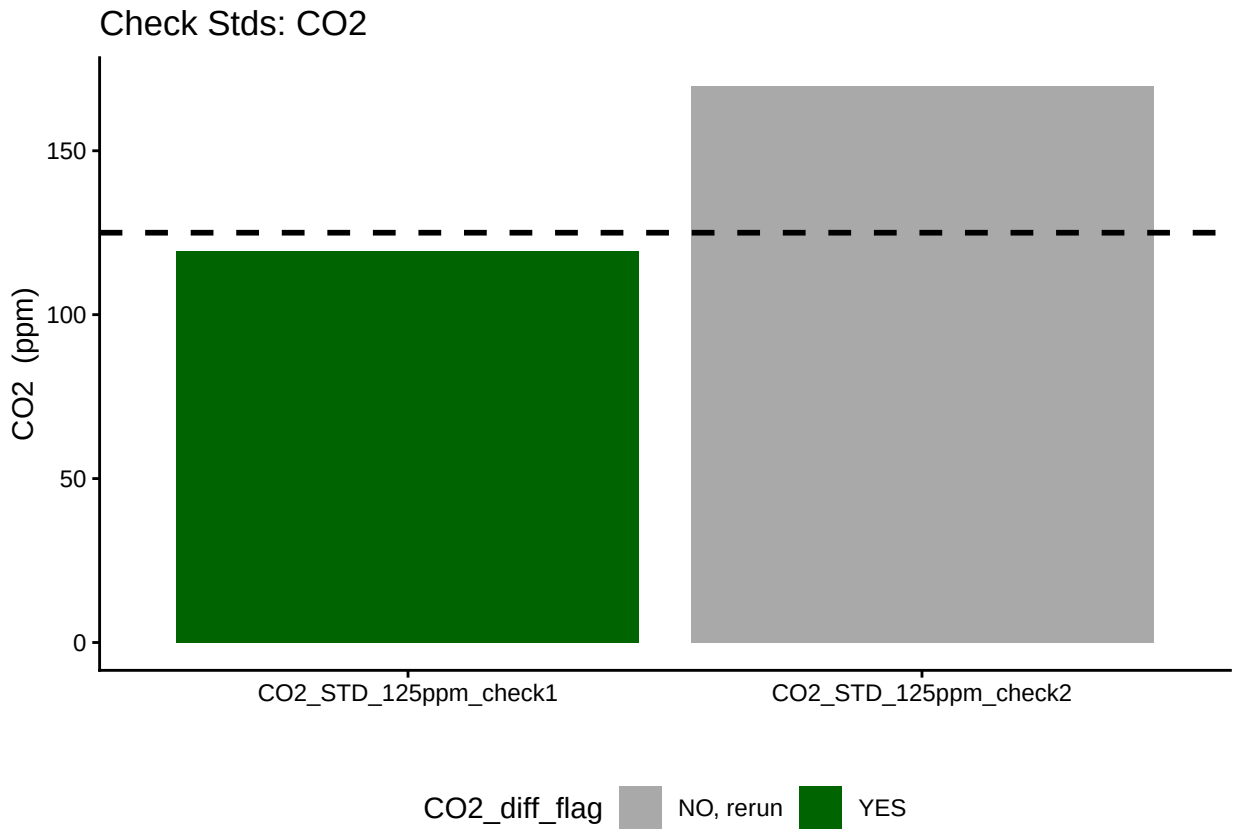
#head(Samples)

##########make flags for any dilutions needed
#highest CH4 standard = 10000
```

```
#highest CO2 standard = 50000

Samples$CH4_Flag <- ifelse(Samples$CH4_Conc_ppm >250, "adl", "Within Range")
Samples$CO2_Flag <- ifelse(Samples$CO2_Conc_ppm >50000, "adl", "Within Range") ##Assuming the CO2 standa
#head(Samples)
```

Check the Check Standards



First look at samples

```
# #multiply the concentration by the dilution factor
# Samples$CH4_Conc_ppm_dilcorr <- (Samples$CH4_Conc_ppm * Samples$Dilution_Factor)
#
# Samples$CO2_Conc_ppm_dilcorr <- (Samples$CO2_Conc_ppm * Samples$Dilution_Factor)

Samples <- Samples %>%
  mutate(CH4_Conc_ppm = ifelse(CH4_Conc_ppm < 0, 0, CH4_Conc_ppm)) %>%
  mutate(CO2_Conc_ppm = ifelse(CO2_Conc_ppm < 0, 0, CO2_Conc_ppm))

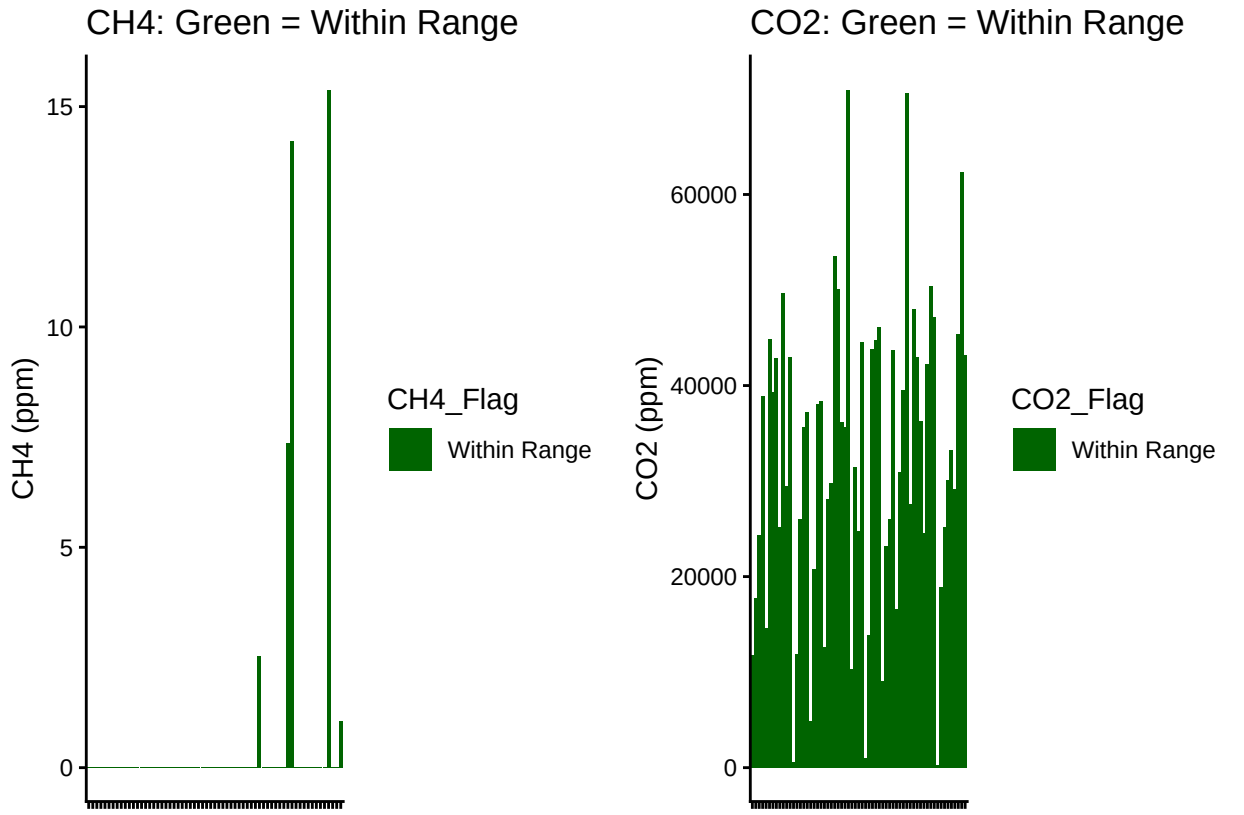

#check results
#head(Samples)

#quick first look at the samples
ch4_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CH4_Conc_ppm, fill=CH4_Flag)) +
```

```
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("darkgreen","red"))+
        theme_classic() +
        labs(x= " ", y="CH4 (ppm)", title="CH4: Green = Within Range") +
        theme(axis.text.x = element_blank())
        # theme(legend.position="none")


co2_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CO2_Conc_ppm, fill=CO2_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c( "darkgreen", "red"))+
        theme_classic() +
        labs(x= " ", y="CO2 (ppm)", title="CO2: Green = Within Range") +
        theme(axis.text.x = element_blank())
        # theme(legend.position="none")

ggarrange(ch4_samples, co2_samples, nrow=1, ncol=2)
```




```
##Percentage of samples that need dilution
Samples %>%
  summarise('Percent CO2 Samples Needing Dilution' = mean(CO2_Flag == "adl", na.rm = TRUE) * 100)
```

```
## # A tibble: 1 x 1
##   `Percent CO2 Samples Needing Dilution`
##                                    <dbl>
```

```
## 1                                     0
```

```
Samples%>%
  summarise('Percent CH4 Samples Needing Dilution' = mean(CH4_Flag == "adl", na.rm = TRUE) * 100)
```

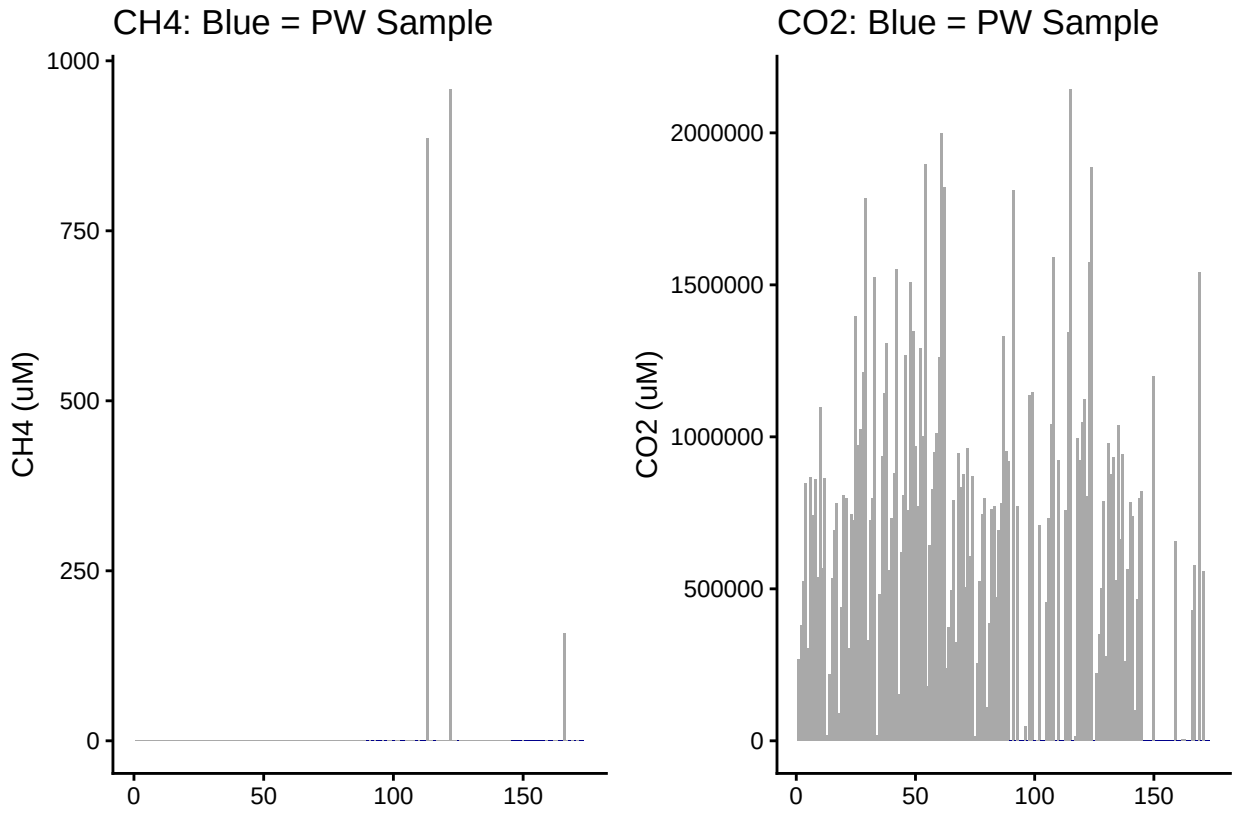
```
## # A tibble: 1 x 1
##   `Percent CH4 Samples Needing Dilution`
##                                    <dbl>
## 1                                      0
```

##Merge samples with metadata

```
metadata <- read_excel("Metadata/TEMPEST_SoilGHG_Scanned_2022.xlsx") %>%
  rename("Grid_Square" = "Grid Square",
         "Sample_Number" = "Sample ID",
         "Sample_Time" = "Time (24hr)",
         "Sample_Type" = "Sample (G/W)",
         "Collection_Date" = "Collection Date",
         "Field_Notes" = "Notes") %>%
  select(-c("CH4 Peak Area", "CO2 Peak Area")) %>%
  mutate(Sample_Time = as.numeric(Sample_Time)) %>%
  mutate(Collection_Date = as.character(Collection_Date)) %>%
  mutate(Timepoint = case_when(
    Collection_Date == "20220620|20220621" ~ NA_character_,
    Collection_Date == "20220622" & Sample_Time < 1000 ~ "T0",
    Collection_Date == "20220622" & Sample_Time > 1000 & Sample_Time < 1400~ "T1",
    TRUE ~ NA_character_
  )) %>%
  mutate(Plot = case_when(
    Plot == "Control" ~ "C",
    Plot == "Freshwater" ~ "FW",
    Plot == "Seawater" ~ "SW",
  ))

Samples_meta <- Samples %>% left_join(metadata, by = c("Plot", "Grid_Square", "Depth", "Collection_Date
```

If samples are water calculate gas in water



Write out processed data & slopes

```
#check results
colnames(Samples_meta)
```

```
##  [1] "Machine"         "Run_Date"        "Sample_ID"       "Plot"
##  [5] "Grid_Square"     "Depth"           "Dilution_Factor" "CO2_Area"
##  [9] "CH4_Area"        "Lab_Notes"       "Collection_Date" "Timepoint"
## [13] "CH4_Curve"       "CH4_Conc_ppm"    "CO2_Conc_ppm"    "CH4_Flag"
## [17] "CO2_Flag"        "Sample_Number"   "Sample_Time"     "Sample_Type"
## [21] "Field_Notes"     "Personnel"       "CH4_Conc_umol"   "CO2_Conc_umol"
```

```
write.csv(Samples_meta, "COMPASS_TEMPEST_SGAS_2022.csv")
```

#end